Ansys Actions

Version: main

Hk migrate fork pr

Migrate fork pr to main repository

```
- name: Migrate fork PR
  uses: ansys/actions/hk-migrate-fork-pr@main
  with:
    bot-email: ${{ secrets.PYANSYS_CI_BOT_EMAIL }}
    bot-username: ${{ secrets.PYANSYS_CI_BOT_USERNAME }}
    comment-id: ${{ github.event.comment.id }}
    github-token: ${{ secrets.PYANSYS_CI_BOT_TOKEN }}
    pr-number: ${{ github.event.issue.number }}
    team-slug: pyansys-maintainers
    user-triggering: ${{ github.event.comment.user.login
}}
```

Ref: Migrate fork PR to main repository

Doc build

Building library documentation

```
- name: Run Ansys documentation building action
  uses: ansys/actions/doc-build@main
```

Ref: Building library documentation

Building library documentation that using xvfb

```
- name: Run Ansys documentation building action
  uses: ansys/actions/doc-build@main
  with:
    requires-xvfb: true
```

Ref: Building library documentation that using XVFB

Installing additional system dependencies for building documentation

```
- name: Run Ansys documentation building action
  uses: ansys/actions/doc-build@main
  with:
    dependencies: graphviz mermaid-cli
```

Ref: Installing additional system dependencies for building documentation

Code style

Running code style checks

```
- name: Run PyAnsys code style checks
  uses: ansys/actions/code-style@main
```

Ref: Running code style checks

Release pypi public

Check vulnerabilities

Check library vulnerabilities

```
- uses: ansys/actions/check-vulnerabilities@main
  with:
    dev-mode: ${{ github.ref != 'refs/heads/main' }}
    python-package-name: ansys-<product>-<library>
    python-version: ${{ env.MAIN_PYTHON_VERSION }}
    token: ${{ secrets.PYANSYS_CI_BOT_TOKEN }}
```

Ref: Check library vulnerabilities

Doc changelog

Create changelog fragment

```
- uses: ansys/actions/doc-changelog@main
  with:
    bot-email: ${{ secrets.PYANSYS_CI_BOT_EMAIL }}
    bot-user: ${{ secrets.PYANSYS_CI_BOT_USERNAME }}
    token: ${{ secrets.PYANSYS_CI_BOT_TOKEN }}
```

Ref: Create changelog fragment

Docker style

Docker style

```
- uses: ansys/actions/docker-style@main
  with:
    directory: docker
    error-level: 1
    recursive: true
```

Ref: Docker style

Hk automerge prs

Automerging dependabot and pre-commit.ci prs

```
- name: Automerging dependabot and pre-commit.ci PRs
  uses: ansys/actions/hk-automerge-prs@main
  with:
    approver: ${{ secrets.PYANSYS_CI_BOT_USERNAME }}
    approver-token: ${{ secrets.PYANSYS_CI_BOT_TOKEN }}
```

Ref: Automerging dependabot and pre-commit.ci PRs

Check pr title

Check pull-request title follows conventional commits

```
- uses: ansys/actions/check-pr-title@main
  with:
    token: ${{ secrets.GITHUB_TOKEN }}
```

Ref: Check pull-request title follows conventional commits

Build ci wheels

Build a c-extension library

```
- name: Build a C-extension library wheel artifacts
  uses: ansys/actions/build-ci-library@main
```

Ref: Build a C-extension library

Check licenses

Check library dependencies ship with valid licenses

```
- name: Validate third party licenses
  uses: ansys/actions/check-licenses@main
  with:
    python-version: ${{ env.MAIN_PYTHON_VERSION }}
```

Ref: Check library dependencies ship with valid licenses

Release pypi test

Build library

Build library basic example

```
- name: Build library source and wheel artifacts
  uses: ansys/actions/build-library@main
  with:
    library-name: ansys-<product>-<library>
```

Ref: Build library basic example

Build wheelhouse

Build wheelhouse for latest python versions

```
- name: Build a wheelhouse of the Python library
  uses: ansys/actions/build-wheelhouse@main
  with:
    library-name: <ansys-product-library>
    operating-system: ${{ matrix.os }}
    python-version: ${{ matrix.python-version }}
```

Ref: Build wheelhouse for latest Python versions

Doc style

Running documentation style checks

```
- name: Running documentation style checks
  uses: ansys/actions/doc-style@main
  with:
    token: ${{ secrets.GITHUB_TOKEN }}
```

Ref: Running documentation style checks

Running documentation style checks on multiple files

```
- name: Install Git and clone project
  uses: actions/checkout@v6
- id: collect-files
  name: Collect files for Vale
  run: "python - <<'PY' >> \"${GITHUB_OUTPUT}\"\nimport
json\nfrom pathlib import\
    \ Path\n\nfiles = sorted(\n    str(path)\n    for path
in Path(\"doc/source\"\
    ).rglob(\"*.rst\")\n    if not
path.is_relative_to(\"doc/source/api\")\n)\nprint(f\"\
    VALE_FILES={json.dumps(files)}\")\nPY\n"
  shell: bash
- name: Running documentation style checks
  uses: ansys/actions/doc-style@main
  with:
    files: ${{ steps.collect-files.outputs.VALE_FILES }}
    token: ${{ secrets.GITHUB_TOKEN }}
```

Ref: Running documentation style checks on multiple files

Hk autorelease

Release from pyproject.toml without lockfile regeneration

```
- name: Auto release
  uses: ansys/actions/hk-autorelease@main
  with:
    bot-email: ${{ secrets.MY_BOT_EMAIL }}
    bot-username: ${{ secrets.MY_BOT_USERNAME }}
    cherry-pick-commits: ${{ inputs.cherry_pick_commits }}
    lock-file-tool: none
    release-type: ${{ inputs.release_type }}
    token: ${{ secrets.MY_BOT_TOKEN }}
    version-location: pyproject.toml
    version-source: pyproject
```

Ref: Release from pyproject.toml without lockfile regeneration

Release from a python module's version

```
- name: Auto release
  uses: ansys/actions/hk-autorelease@main
  with:
    bot-email: ${{ secrets.MY_BOT_EMAIL }}
    bot-username: ${{ secrets.MY_BOT_USERNAME }}
    cherry-pick-commits: ${{ inputs.cherry_pick_commits }}
    release-type: ${{ inputs.release_type }}
    token: ${{ secrets.MY_BOT_TOKEN }}
    version-location: src/ansys/example/_version.py
    version-source: python-module
```

Ref: Release from a Python module's **version**

Release regenerating a poetry lockfile

```
- name: Release and regenerate poetry.lock
  uses: ansys/actions/hk-autorelease@main
  with:
    bot-email: ${{ secrets.MY_BOT_EMAIL }}
    bot-username: ${{ secrets.MY_BOT_USERNAME }}
    cherry-pick-commits: ${{ inputs.cherry_pick_commits }}
    lock-file-tool: poetry
    release-type: ${{ inputs.release_type }}
    token: ${{ secrets.MY_BOT_TOKEN }}
    version-location: pyproject.toml
    version-source: pyproject
```

Ref: Release regenerating a Poetry lockfile

Release from a version file

```
- name: Auto release
  uses: ansys/actions/hk-autorelease@main
  with:
    bot-email: ${{ secrets.MY_BOT_EMAIL }}
```

```
    bot-username: ${{ secrets.MY_BOT_USERNAME }}
    cherry-pick-commits: ${{ inputs.cherry_pick_commits }}
    release-type: ${{ inputs.release_type }}
    token: ${{ secrets.MY_BOT_TOKEN }}
    version-location: ansys/api/geometry/VERSION
    version-source: file
```

Ref: Release from a VERSION file

Release regenerating a uv lockfile in a subdirectory

```
- name: Release and regenerate packages/example/uv.lock
  uses: ansys/actions/hk-autorelease@main
  with:
    bot-email: ${{ secrets.MY_BOT_EMAIL }}
    bot-username: ${{ secrets.MY_BOT_USERNAME }}
    cherry-pick-commits: ${{ inputs.cherry_pick_commits }}
    lock-file-tool: uv
    release-type: ${{ inputs.release_type }}
    token: ${{ secrets.MY_BOT_TOKEN }}
    version-location: packages/example/pyproject.toml
    version-source: pyproject
```

Ref: Release regenerating a uv lockfile in a subdirectory

Release from setup.py

```
- name: Auto release
  uses: ansys/actions/hk-autorelease@main
  with:
    bot-email: ${{ secrets.MY_BOT_EMAIL }}
    bot-username: ${{ secrets.MY_BOT_USERNAME }}
    cherry-pick-commits: ${{ inputs.cherry_pick_commits }}
    release-type: ${{ inputs.release_type }}
    token: ${{ secrets.MY_BOT_TOKEN }}
    version-location: setup.py
    version-source: setup-py
```

Ref: Release from setup.py

Doc deploy changelog

Update changelog for new tag

```
- uses: ansys/actions/doc-deploy-changelog@main
  with:
    bot-email: ${{ secrets.PYANSYS_CI_BOT_EMAIL }}
    bot-user: ${{ secrets.PYANSYS_CI_BOT_USERNAME }}
    token: ${{ secrets.PYANSYS_CI_BOT_TOKEN }}
```

Ref: Update CHANGELOG for new tag

Doc deploy stable

Deploy stable documentation

```
- name: Deploy the stable documentation
  uses: ansys/actions/doc-deploy-stable@main
  with:
    bot-email: ${{ secrets.PYANSYS_CI_BOT_EMAIL }}
    bot-user: ${{ secrets.PYANSYS_CI_BOT_USERNAME }}
    cname: <library>.docs.pyansys.com
    token: ${{ secrets.GITHUB_TOKEN }}
```

Ref: Deploy stable documentation

Doc deploy custom path

Deploy documentation to a custom path

```
- name: Deploy the documentation to a custom subdirectory
  uses: ansys/actions/doc-deploy-custom-path@main
  with:
    bot-email: ${{ secrets.PYANSYS_CI_BOT_EMAIL }}
    bot-user: ${{ secrets.PYANSYS_CI_BOT_USERNAME }}
    cname: <library>.docs.pyansys.com
    custom-path: nightly
    token: ${{ secrets.GITHUB_TOKEN }}
```

Ref: Deploy documentation to a custom path

Hk tag repository version

Tag repository version

```
- uses: actions/checkout@v6
  with:
    fetch-depth: 0
    fetch-tags: true
- name: Update latest vX and vX.Y tag
  uses: ansys/actions/hk-tag-repository-version@main
  with:
    gh-token: ${{ secrets.GITHUB_TOKEN }}
    release-version: ${{ github.event.release.tag_name }}
```

Ref: Tag repository version

Tag repository version

```
- uses: actions/checkout@v6
  with:
    fetch-depth: 0
    fetch-tags: true
- env:
    SEMANTIC_VERSION: ${{ github.event.release.tag_name }}
  id: major_minor_version
  name: vX.Y version from tag
  run: 'MAJOR_VERSION=$(echo ${SEMANTIC_VERSION} |  cut -d
''v'' -f 2 | cut -d ''.''
    -f 1)

    MINOR_VERSION=$(echo ${SEMANTIC_VERSION} |  cut -d
''v'' -f 2 | cut -d ''.'' -f
    2)

    echo
"MAJOR_MINOR_VERSION=v${MAJOR_VERSION}.${MINOR_VERSION}"
>> $GITHUB_OUTPUT

    '
  shell: bash
- name: Update latest vX and vX.Y tag
  uses: ansys/actions/hk-tag-repository-version@main
  with:
    gh-token: ${{ secrets.GITHUB_TOKEN }}
    latest-branch-name: release/${{
steps.major_minor_version.outputs.MAJOR_MINOR_VERSION
      }}
```

Ref: Tag repository version

Release github

Release to github

```
- name: Release to GitHub
  uses: ansys/actions/release-github@main
  with:
    library-name: ansys-<product>-<library>
```

Ref: Release to GitHub

Check actions security

Check actions security

```
- uses: ansys/actions/check-actions-security@main
  with:
    auditing-level: high
    generate-summary: true
    token: ${{ secrets.GITHUB_TOKEN }}
    trust-ansys-actions: true
```

Ref: Check actions security

Release pypi private

Release to private pypi

```
- name: Release to the private PyPI repository
  uses: ansys/actions/release-pypi-private@main
  with:
    library-name: ansys-<product>-<library>
    twine-token: ${{ secrets.PYANSYS_PYPI_PRIVATE_PAT }}
    twine-username: __token__
```

Ref: Release to private PyPI

Hk package clean except

Perform versions cleanup - except certain tags

```
- name: Perform versions cleanup - except certain tags
  uses: ansys/actions/hk-package-clean-except@main
  with:
    package-name: mypackage
    tags-kept: latest, latest-unstable
    token: ${{ secrets.GITHUB_TOKEN }}
```

Ref: Perform versions cleanup - except certain tags

Hk package clean untagged

Perform untagged versions cleanup

```
- name: Perform untagged versions cleanup
  uses: ansys/actions/hk-package-clean-untagged@main
  with:
    package-name: mypackage
    token: ${{ secrets.GITHUB_TOKEN }}
```

Ref: Perform untagged versions cleanup

Doc deploy dev

Deploy developers documentation

```
- name: Deploy the latest documentation
  uses: ansys/actions/doc-deploy-dev@main
  with:
    bot-email: ${{ secrets.PYANSYS_CI_BOT_EMAIL }}
    bot-user: ${{ secrets.PYANSYS_CI_BOT_USERNAME }}
    cname: <library>.docs.pyansys.com
    token: ${{ secrets.GITHUB_TOKEN }}
```

Ref: Deploy developers documentation

Doc deploy pr

Deploy pr documentation

```
- uses: ansys/actions/doc-deploy-pr@main
  with:
    bot-email: ${{ secrets.PYANSYS_CI_BOT_EMAIL }}
    bot-user: ${{ secrets.PYANSYS_CI_BOT_USERNAME }}
    cname: ${{ env.DOCUMENTATION_CNAME }}
    maximum-pr-doc-deployments: 3
    token: ${{ secrets.GITHUB_TOKEN }}
```

Ref: Deploy PR documentation

Tests pytest

Testing library with different operating systems and python versions

```
- name: Run pytest
  uses: ansys/actions/tests-pytest@main
  with:
    pytest-extra-args: --cov=ansys.<library>
--cov-report=term --cov-report=html:.cov/html
    pytest-markers: -k 'mocked'
```

Ref: Testing library with different operating systems and Python versions

Optimized testing with different operating systems and python versions

```
- name: Login into the GitHub container registry
  uses: docker/lo-
gin-action@f4ef78c080cd8ba55a85445d5b36e214a81df20a
  with:
    password: ${{ secrets.GITHUB_TOKEN }}
    registry: ghcr.io
    username: ${{ secrets.GH_USERNAME }}
- env:
    LICENSE_SERVER: ${{ secrets.LICENSE_SERVER }}
  name: Pull, launch and verify desired Ansys service
  run: '...'
- name: Run pytest with desired markers and extra
arguments
  uses: ansys/actions/tests-pytest@main
  with:
    pytest-extra-args: ${{ matrix.cfg.extra-args }}
    pytest-markers: ${{ matrix.cfg.markers }}
    requires-xvfb: true
```

Ref: Optimized testing with different operating systems and Python versions